

myElectron Solar IoT

Open-Source ESP32 Solar-Powered IoT Development Platform

Integrated solar charging and **18650 battery** power management

Hardware-based **ATtiny13A** power cycling for longevity

LoRaWAN and **4G-LTE** modular expansion

IP65 enclosure readiness for outdoor use



Hardware Features, Interfaces & Applications

A complete solar-powered IoT platform for robust deployment and versatile development

BUILT-IN HARDWARE

ESP32-WROOM 32 MCU
CN3065 solar charger
& 5V boost converter
ATTiny13A hard power
cycle controller
Tilt switch, status LED,
reset/program switches,
waterproof rocker switch

EXPANSION INTERFACES

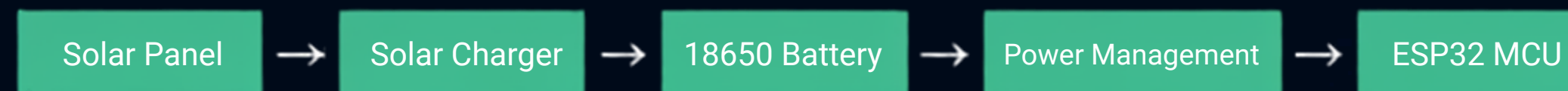
GPIO & sensor header
SCL, SDA, i034, i035,
i014, i027, 3.3V, 5V, GND

LoRaWAN & 4G-LTE
modular headers for
flexible connectivity

ECOSYSTEM COMPATIBILITY

myElectron solar panels
18650 batteries
LoRaWAN modules
Sensor probes
IP65 enclosures

POWER FLOW



REAL-WORLD APPLICATIONS

Smart agriculture
Environmental & air
quality monitoring

Remote telemetry &
weather stations

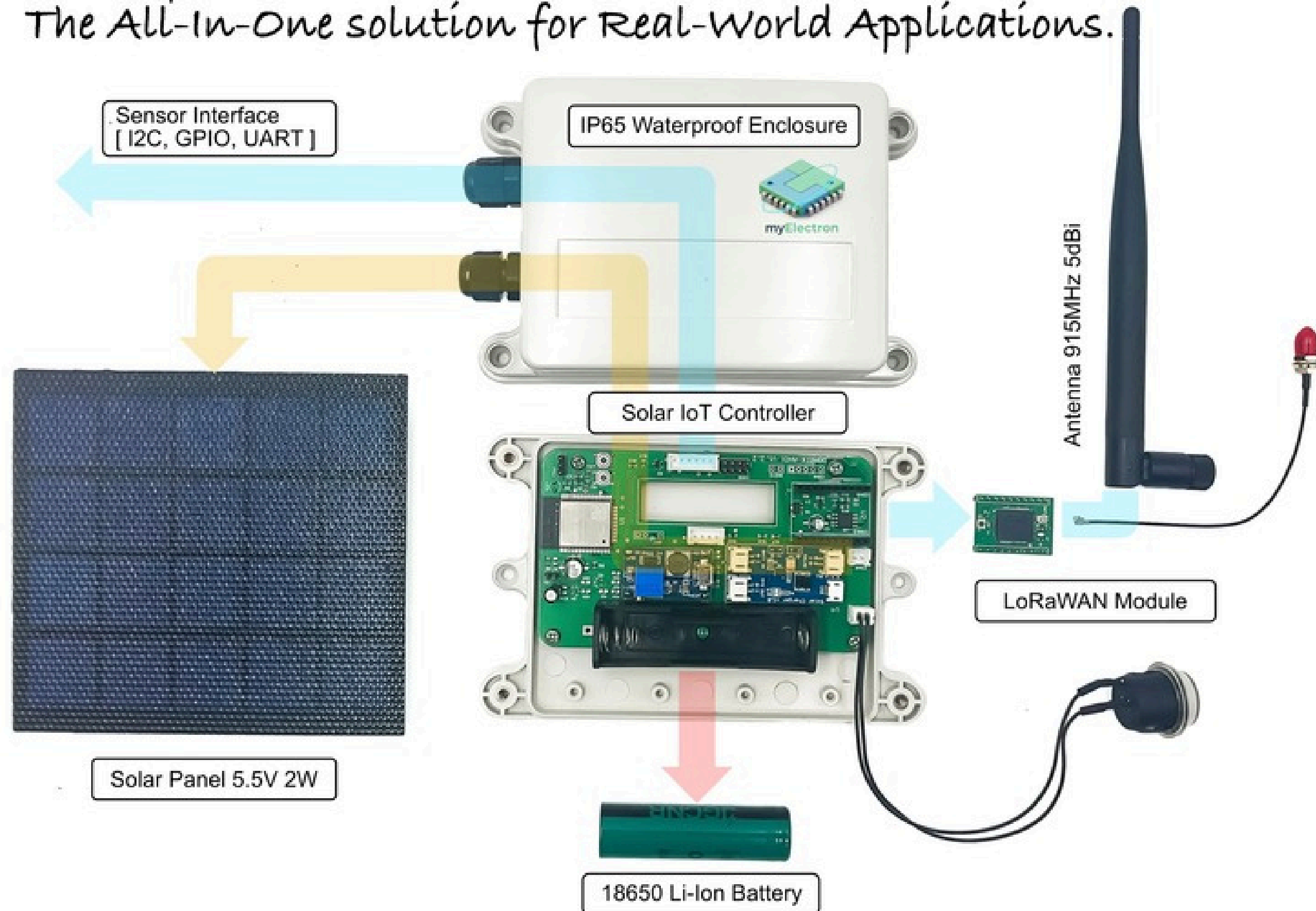
Off-grid LoRaWAN/
cellular sensor nodes

Embedded systems
development &
prototyping



SOLAR IoT : ESP32 DEVELOPMENT KIT

Build Your Own Solar-Powered IoT Node.
The All-In-One solution for Real-World Applications.





UART Interface
for debug/Program/
Serial Device Interface

ESP32 WROOM-32
Wi-Fi/BLE+4Mb Flash

Integrated TILT Switch
apply for wireless OTA/
Tampered Proof function

5V DC-DC Step-Up
Boost Converter

18650 Rechargeable
Li-Ion Battery Holder

I2C / GPIO / 3.3V / 5V
for Sensor Interface.

LoRa / 4G-LTE
Expansion Interface

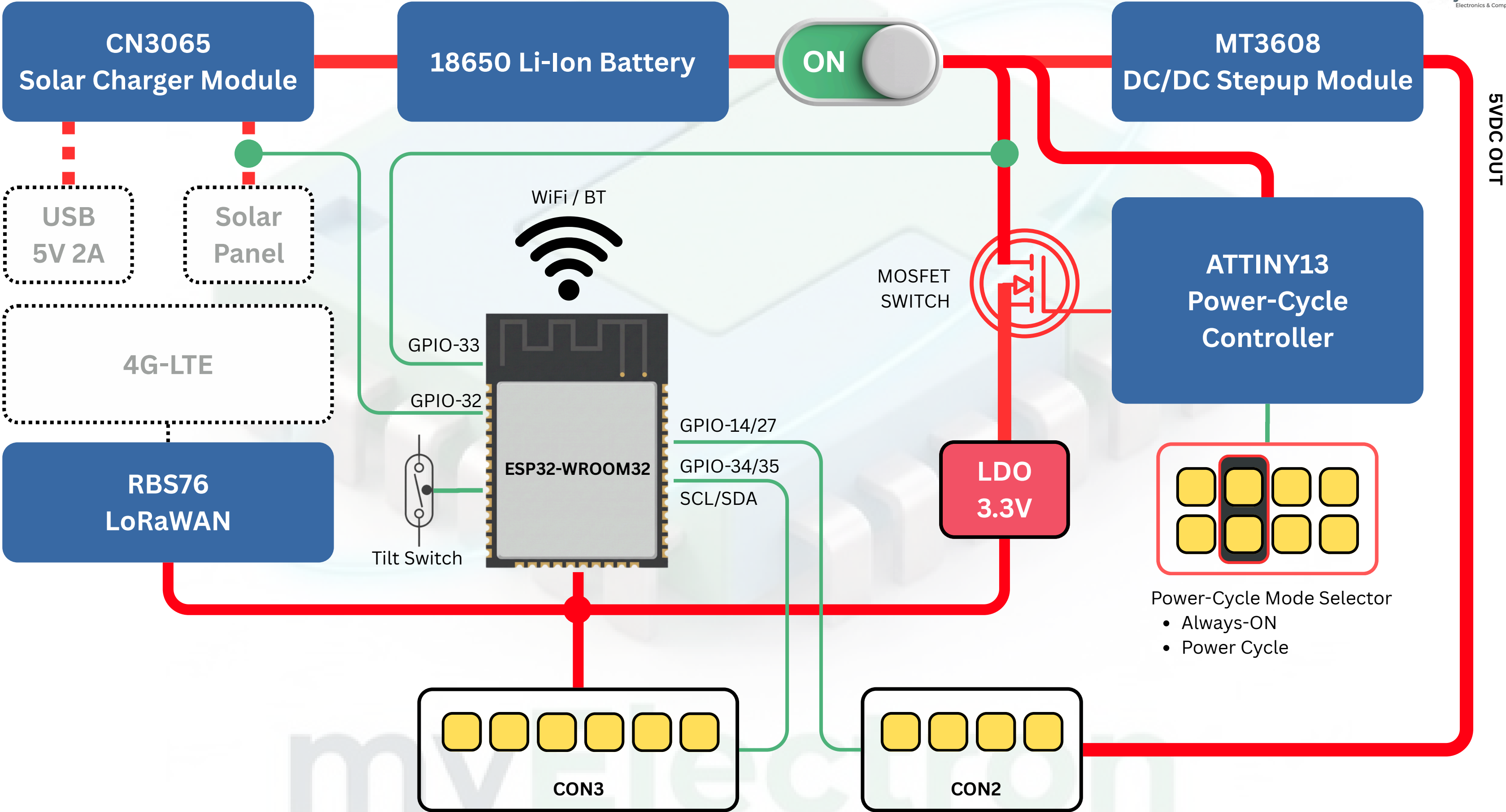
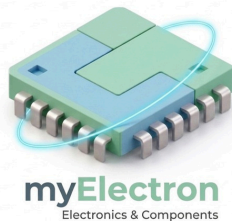
ATTINY13A : Supervisory
Power Management
Automatic Power Cycling

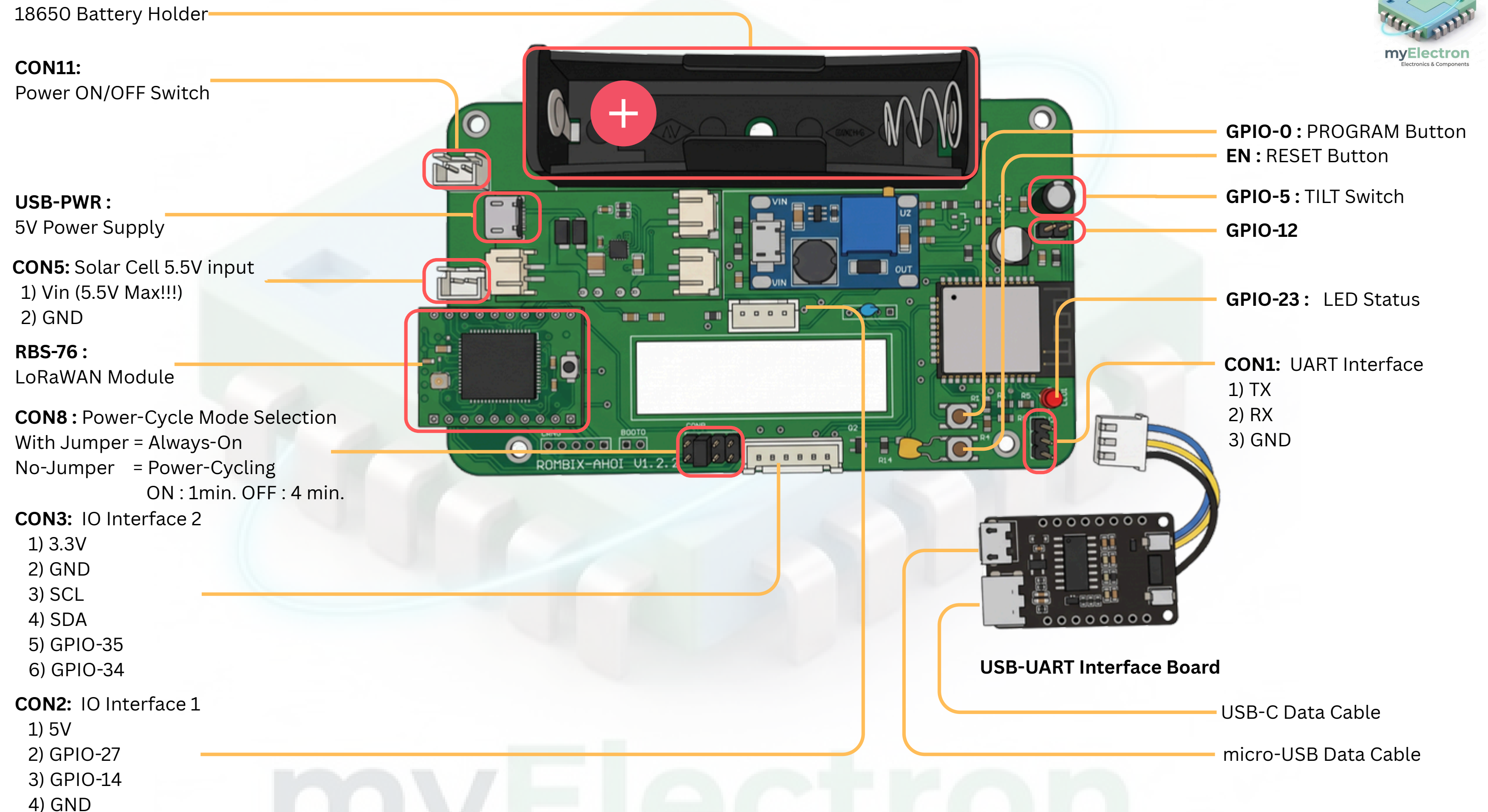
Integrated Solar Charger
Adaptive Charging Control

Waterproof
Power On/Off Switch

myElectron : Solar IoT Controller

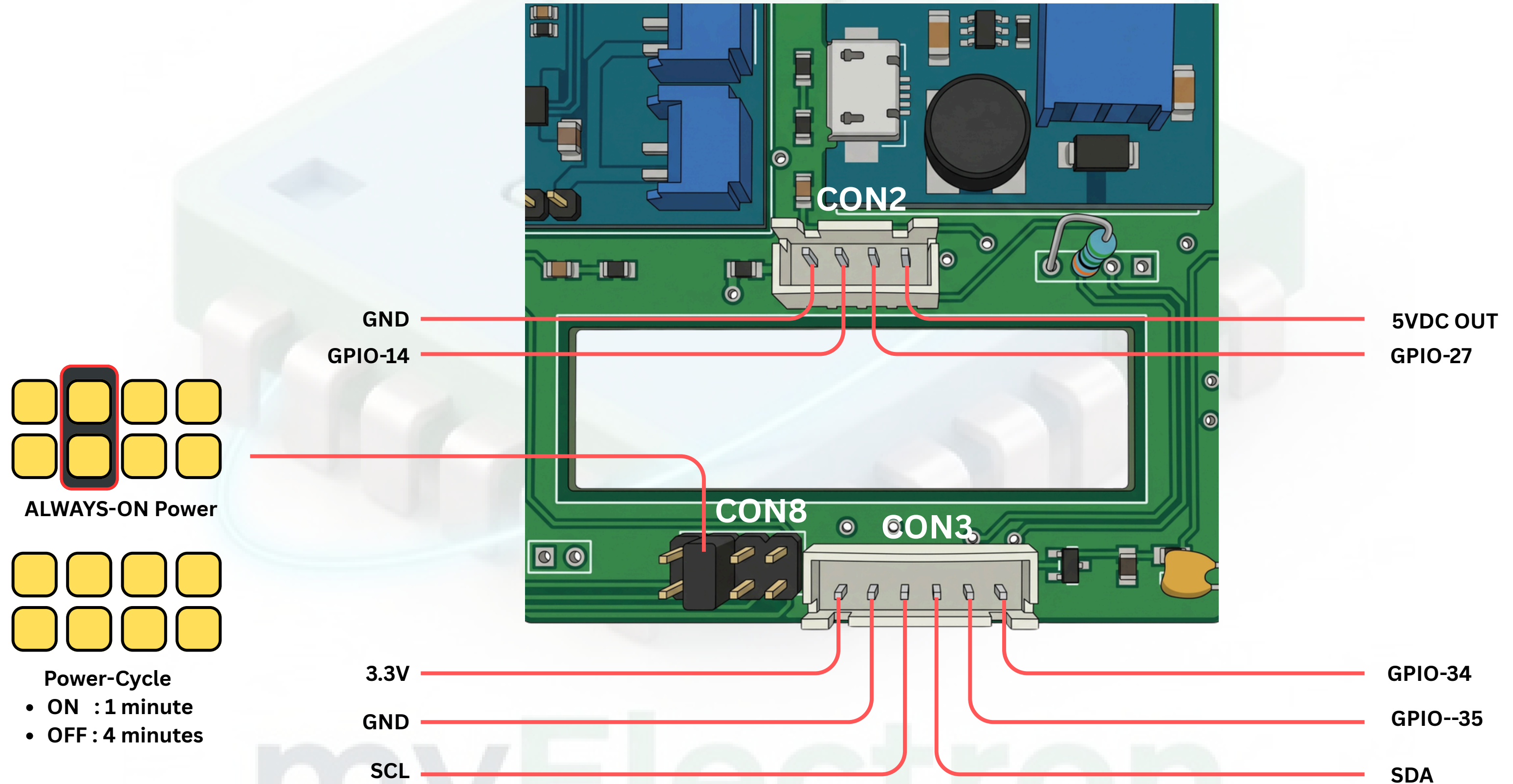
myElectron Solar-IoT Controller Block Diagram



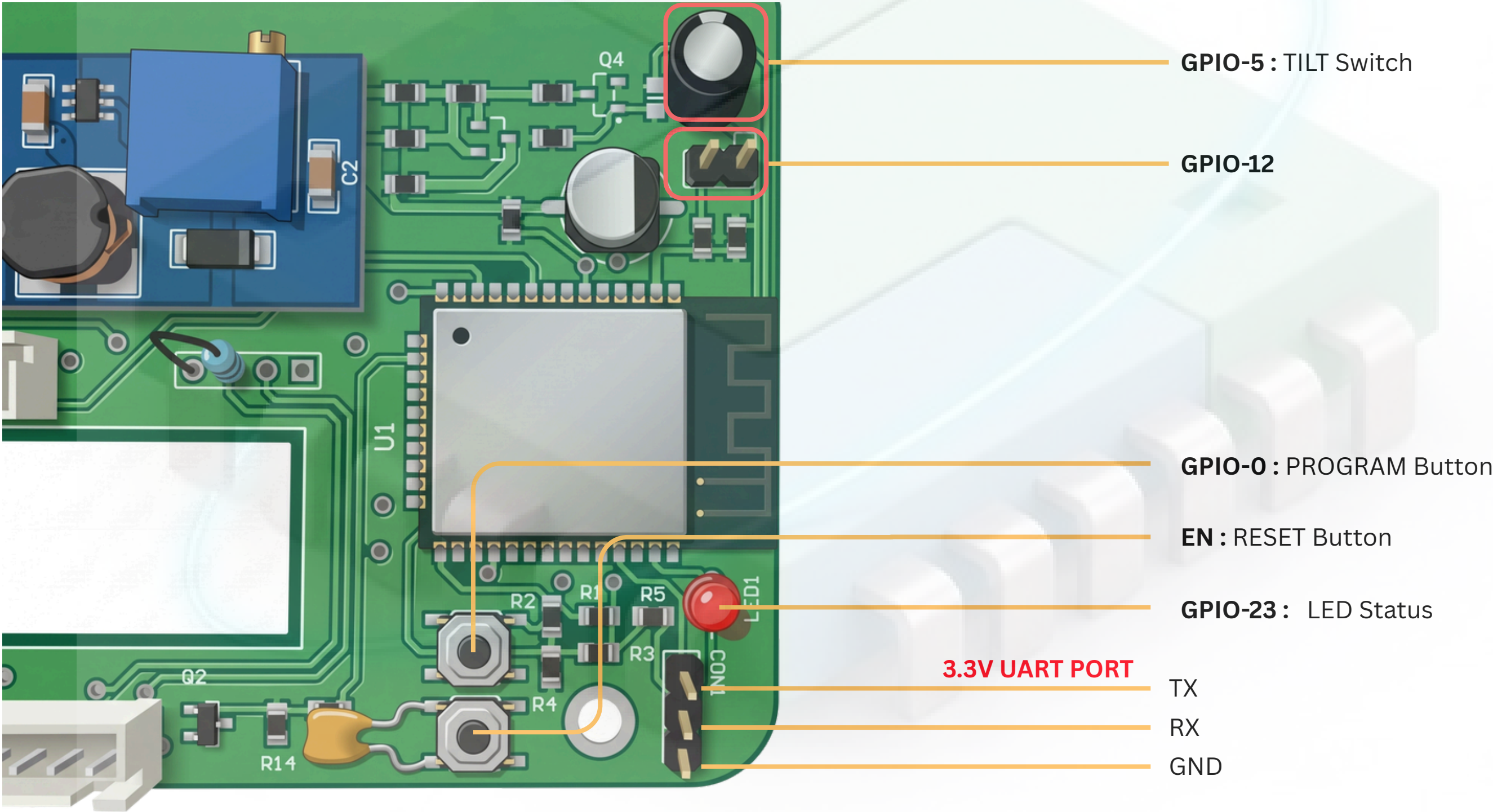
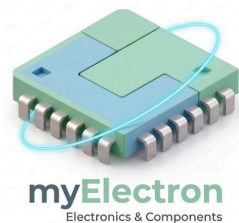


myElectron Solar-IoT Controller Pinout Diagram

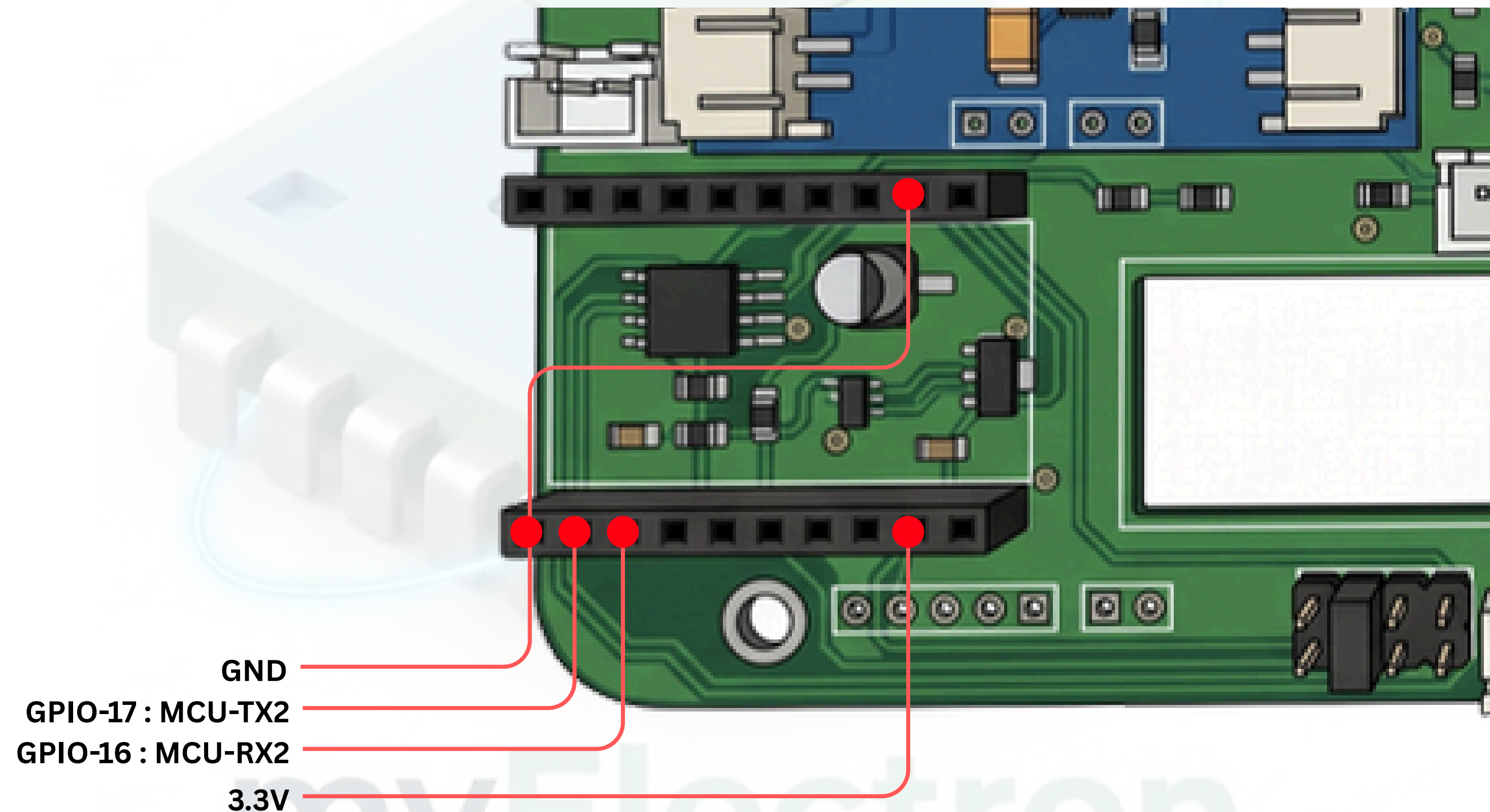
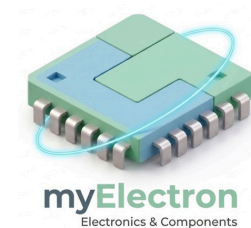
myElectron Solar-IoT Controller GPIO Pinout Diagram



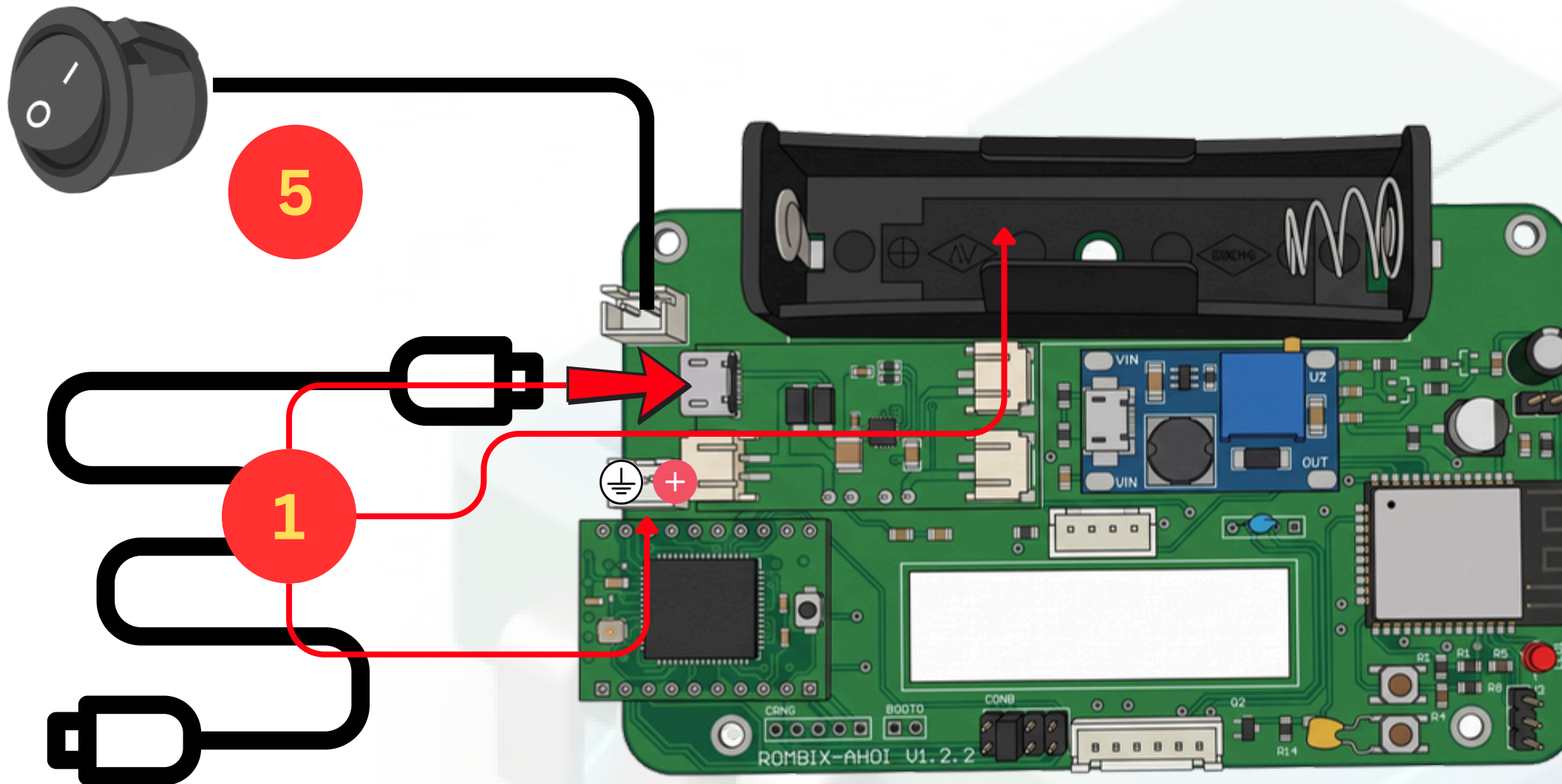
myElectron Solar-IoT Controller ESP32 PG/RES/UART Pinout Diagram



myElectron Solar-IoT Controller: Communication Socket Pinout Diagram



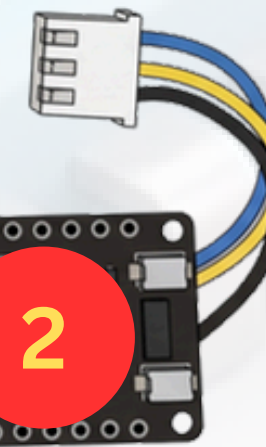
GET STARTED →



- 1) Connect a 5V power supply using a USB power cable (recommended: 5V, 2A).
 - > Alternatively, insert an 18650 Li-ion battery into the battery holder.
 - > You may also supply 5V DC through CON11 (the solar-cell power port).

*** Caution: Maximum voltage is 5.5V. ***

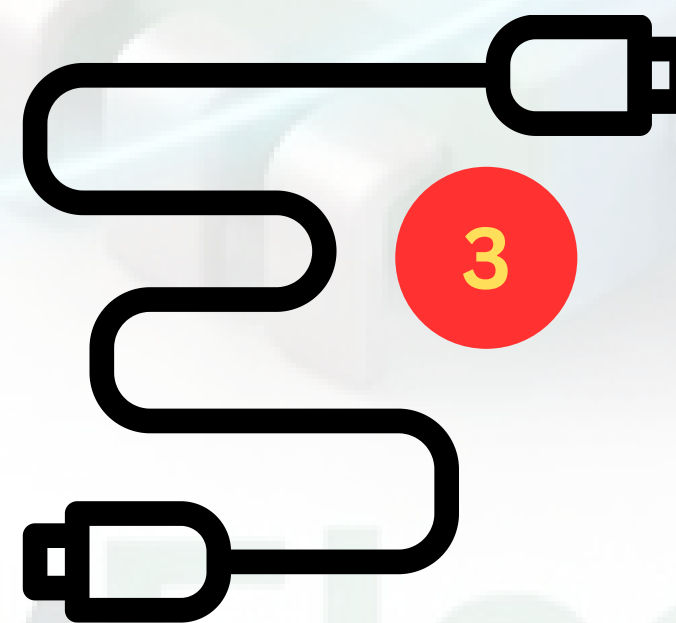
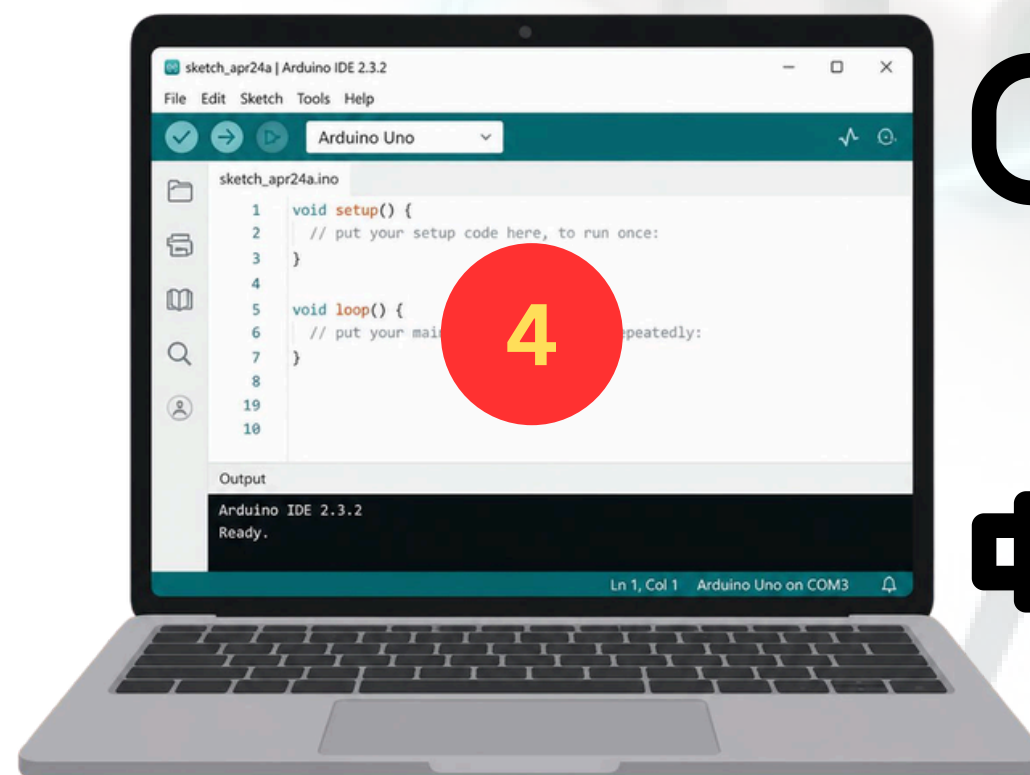
- 2) Connect the USB-to-UART interface board. Connect the black cable to GND.



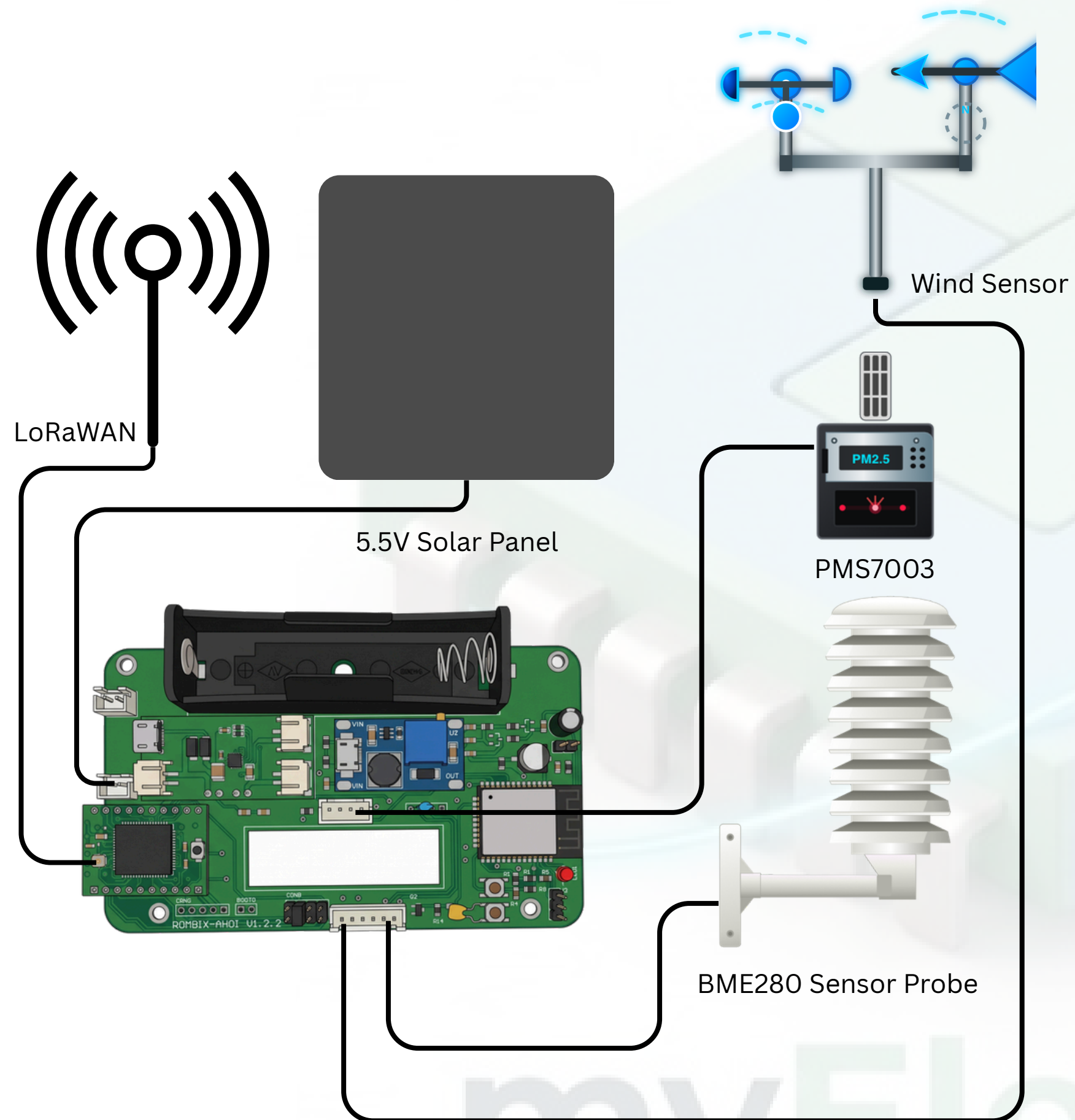
- 3) Connect the USB data cable to your PC and configure the serial port as follows:
115200 baud, 8 data bits, no parity, 1 stop bit (115200/8/N/1).

- 4) Open the serial port.

- 5) Power on the Solar-IoT board.



myElectron Solar-IoT Controller Getting Start



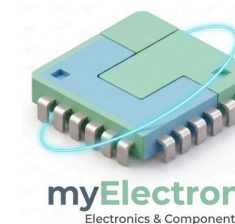
What is Factory Self-Test

The Solar-IoT Controller will be preloaded at the factory with a system test program. User can record the serial number and then overwrite the existing program. The following will be tested:

- Read the chip ID (used as the board's serial number)
- Onboard battery voltage ADC
- Onboard solar PV voltage ADC
- Tilt switch status
- I2C (connected to the BME280 sensor probe)
- Pulse counter (no value, the sensor not connected)
- PMS7003 laser dust sensor (no value, the sensor not connected)
- Test communication with the LoRaWAN module



What is showing on Factory Self-Test Screen



AQI-AH01 V1.101.17

ChipID : XXXXXXXXXX *** This is your Serial No. Please noted. ***

LOOP :1

LoRa SLEEP >> sleep 900 sec uart_on

SETTING Vref...

VB = 0

OK

ADC 1425

BATT 3.66 V

Scantime = 22000

H1-max = 65535, H1-min = 65535

H2-max = 65535, H2-min = 65535

ID : XXXXXXXXXX

SN : XXXXXXXXXX

TX : 7

Init DTHP... OK!

*** System initial for factory self-test***

*** Initial I2C for BME280 sensor probe ***

Vref 1895

ADC 1425

BATT 4.35 V

*** Checking battery ADC on board ***

TILT: 0

*** Checking Tilt Switch Status ***

ADC 12

PV 0.04 V

*** Checking Solar PV voltage ***

T: 30.48 °C H: 66.22 %RH P: 100876.98 Pa

T: 30.47 °C H: 66.09 %RH P: 100876.74 Pa

T: 30.47 °C H: 66.09 %RH P: 100876.58 Pa

*** Checking BME280 ***

AN_H1_Max: 65535 AN_H1_Min: 65535 AN_H1_value: 0

AN_H2_Max: 65535 AN_H2_Min: 65535 AN_H2_value: 2245

Pulse : 1 RPM : 0.00 Speed : 0.00 m/s

Read-Time : [5219]

*** Checking Pulse counter ***

- Not connected here

Using xPM01= 0

Using xPM25= 0

Using xPM10= 0

Using Last PM1 = 0 PM2.5 = 0 PM10 = 0

*** Checking PMS7003 ***

- Not connected here

MSG = 4b01b300040be719d199ecea000.....

Initial LoRaWAN...

DEBUG-LOG >> Ok

SET ADR-OFF >> Ok

SET DR1 >> Ok

>> Ok

SET TX-CNF >> Ok

MAX-EIRP 7

>> Ok

>> Ok

DevEUI: 990990FEAA194806>> Ok

DevADDR: AA194806>> Ok

>> Ok

Join >> accepted

Sending...

*** Checking RBS76-LoRaWAN Module ***

- Only signal test, no gateway connected.

....

....

....

LoRa SLEEP >> sleep 900 sec uart_on

>> err

Sending 3 time in 19s.

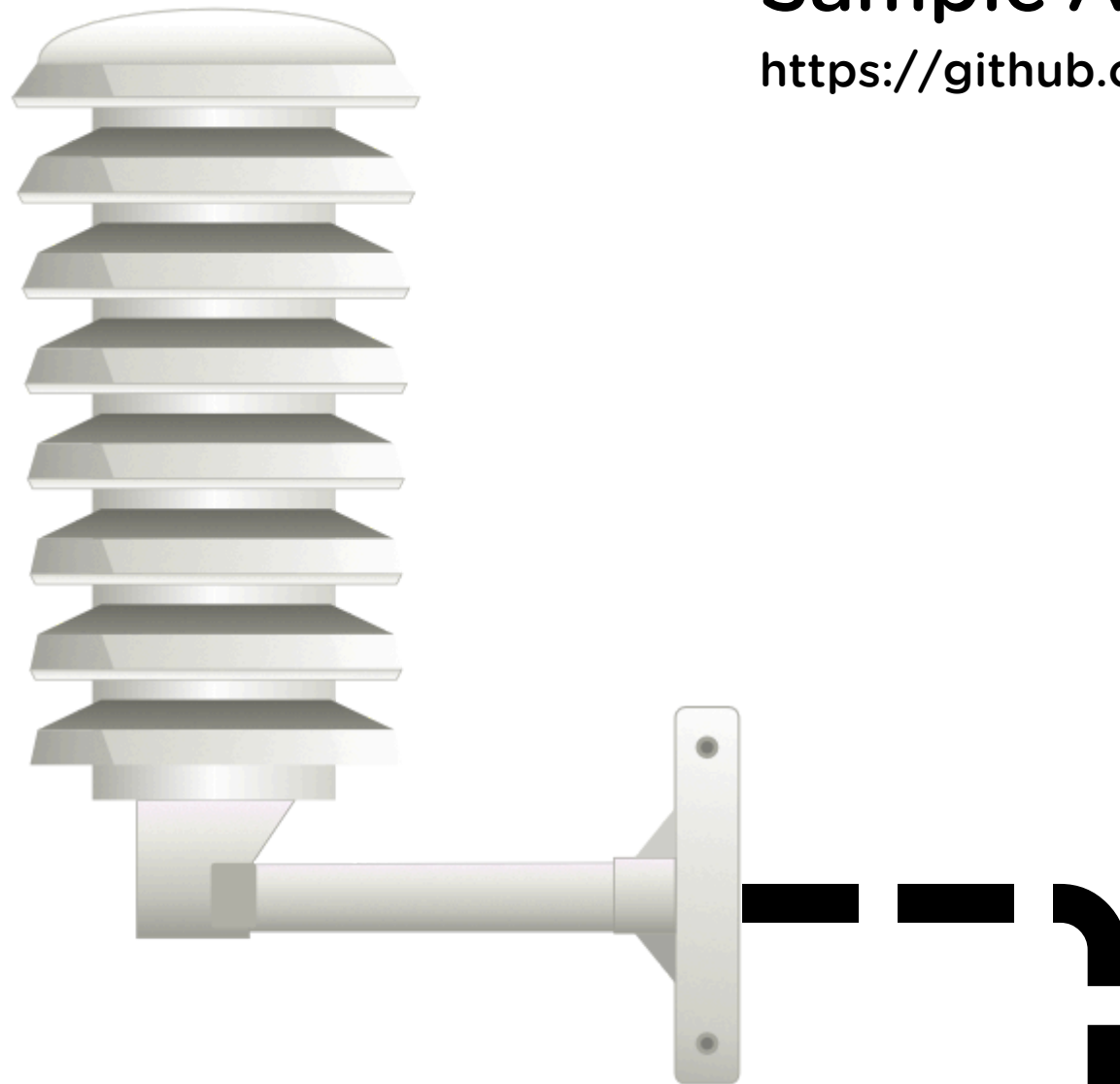
On-Time = 34 sec.

SLEEP 400 sec.

*** Enter Deep-Sleep ***

Sample Application : BME280 Interface

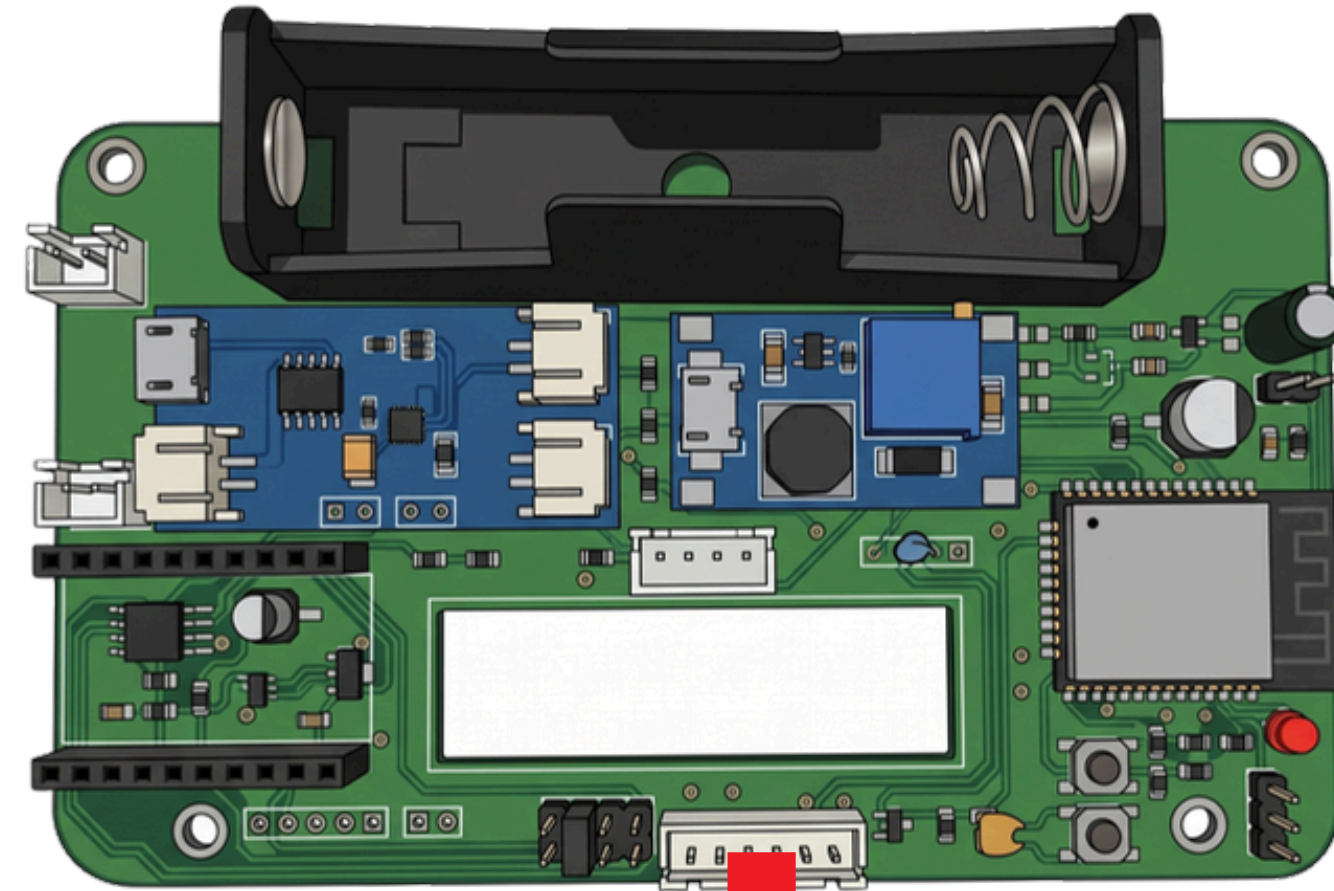
https://github.com/iot2home-myelectron/solar_iot_devkit/tree/Technical-Document/code/bme280



i-BME280 in Solar Radiation Shield
• Best for outdoor application



i-BME280 in PE Protection Shield
• Best for Smartfarm application



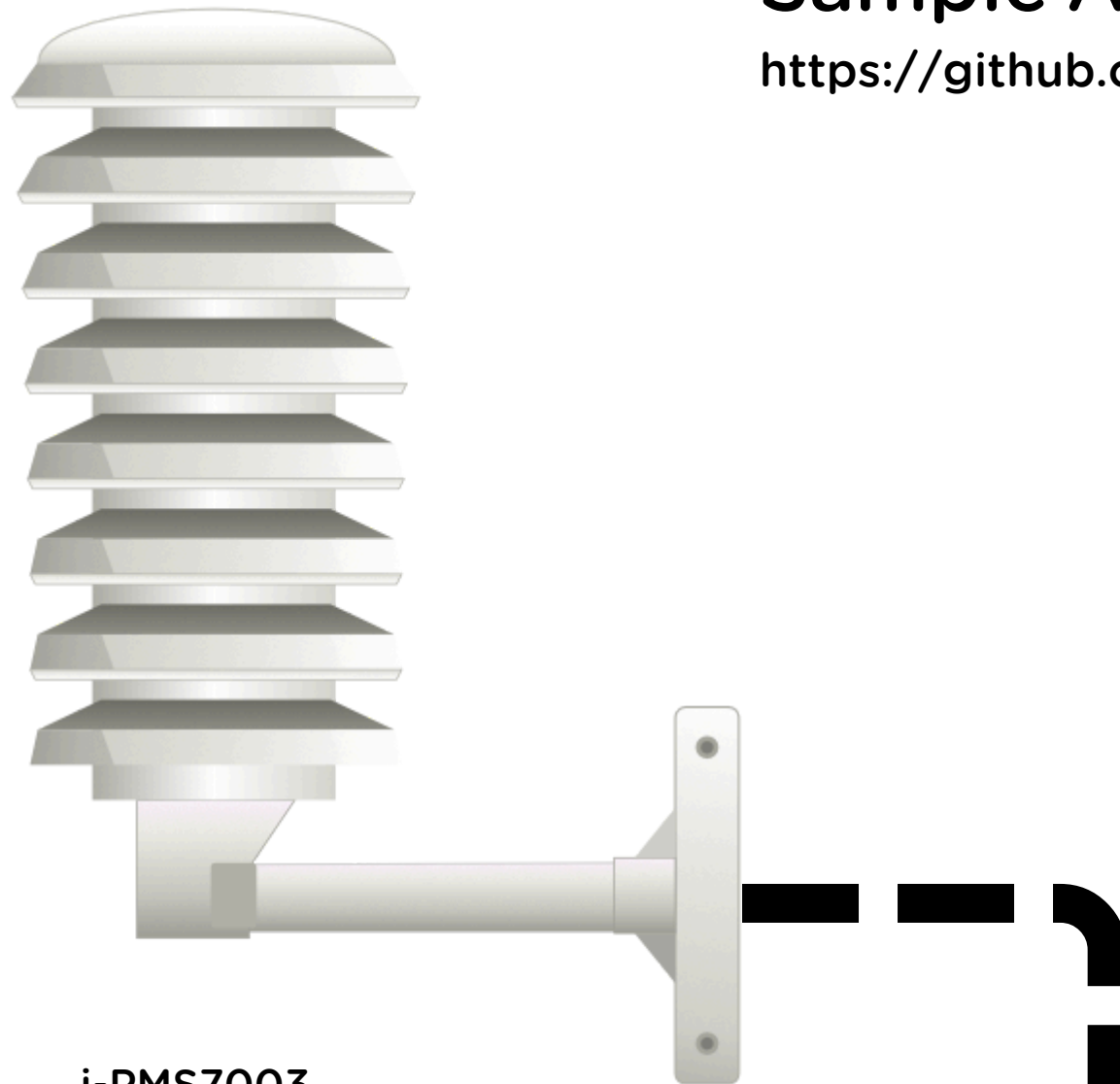
CON3

3.3V GND SCL SDA



Sample Application : PMS7003 Interface

https://github.com/iot2home-myelectron/solar_iot_devkit/tree/Technical-Document/code/pms7003

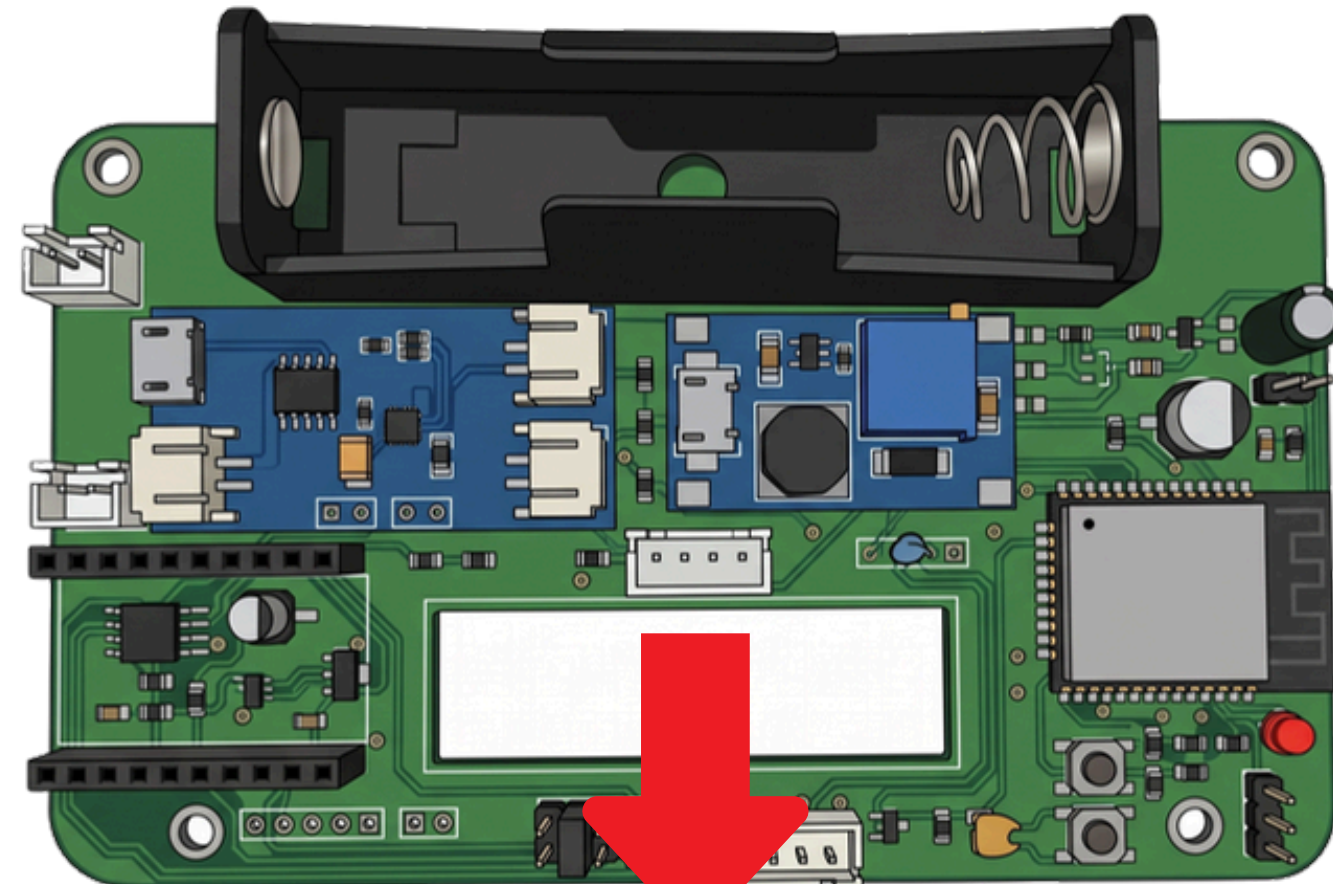


i-PMS7003

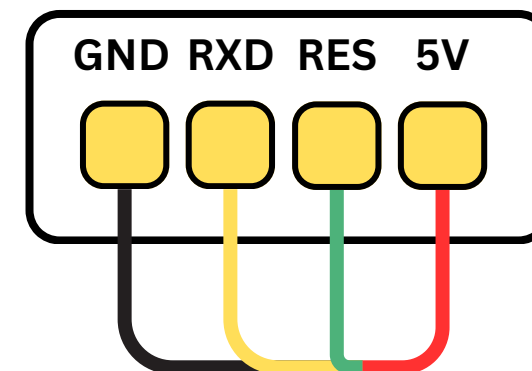
- Integrated in Solar Radiation Shield
- Best for outdoor application



PMS7003 Module



CON2



Pin1	-----	VCC
Pin2	-----	VCC
Pin3	-----	GND
Pin4	-----	GND
Pin5	-----	Reset
Pin6	-----	N/C
Pin7	-----	RX
Pin8	-----	N/C
Pin9	-----	TX
Pin10	-----	Set